10. Simulation and Performance Evaluation of NOMA and Successive Interference Cancellation.

Objectives of This Experiment

1. To model a two-user power-domain NOMA transmission system using MATLAB.

2. To implement successive interference cancellation (SIC) at the receiver for separating the superimposed user signals.

3. To simulate and compare user throughput and/or BER for selected power-allocation settings.

4. To compare the simplified NOMA performance with an equivalent OFDMA transmission case and explain the effect of orthogonal and non-orthogonal resource allocation.

Expected Outcome from This Experiment

• Develop a MATLAB model for a two-user power-domain NOMA system.

• Implement power-domain superposition transmission and SIC-based signal detection.

• Evaluate BER and/or throughput for different power-allocation coefficients.

• Compare NOMA and OFDMA results and interpret the advantages and limitations of non-orthogonal transmission.

Software Required: MATLAB

Program:

```
clc;
clear;
close all;

%% 1. Input Parameters Setup
```

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N_bits = 50000;
SNR_dB = 0:2:20;

alpha1 = 0.75;      % Far user power allocation
alpha2 = 1 - alpha1; % Near user power allocation

ber_User1 = zeros(length(SNR_dB), 1);
ber_User2 = zeros(length(SNR_dB), 1);
ber_OFDMA = zeros(length(SNR_dB), 1);

%% 2. Data Processing Execution Loops
for s = 1:length(SNR_dB)

   snr_linear = 10^(SNR_dB(s)/10);
   noise_variance = 1/snr_linear;

   % Source Bits Generation
   bits_u1 = randi([0 1], N_bits, 1);
   bits_u2 = randi([0 1], N_bits, 1);

   % BPSK Constellation Mapping
   x1 = 2*bits_u1 - 1;
   x2 = 2*bits_u2 - 1;

   % Power-Domain NOMA Superposition
   x_noma = sqrt(alpha1)*x1 + sqrt(alpha2)*x2;

   % AWGN Channel
```

```
noise = sqrt(noise_variance/2) * ...
   (randn(N_bits,1) + 1i*randn(N_bits,1));

y_noma = real(x_noma + noise);

%% 3. Processing User 1 (Far User)
dec_u1 = y_noma > 0;

est_x1 = 2*double(dec_u1) - 1;

ber_User1(s) = sum(bits_u1 ~= dec_u1)/N_bits;

%% 4. Successive Interference Cancellation (SIC)
y_sic = y_noma - sqrt(alpha1)*est_x1;

dec_u2 = y_sic > 0;

ber_User2(s) = sum(bits_u2 ~= dec_u2)/N_bits;

%% 5. OFDMA Benchmark Processing
noise_ofdma = sqrt(noise_variance/2)*randn(N_bits,1);

y_ofdma = x1 + noise_ofdma;

dec_ofdma = y_ofdma > 0;

ber_OFDMA(s) = sum(bits_u1 ~= dec_ofdma)/N_bits;
```

```
end

%% 6. Performance Graph Rendering
figure('Name','NOMA SIC Evaluation Matrix');

semilogy(SNR_dB, ber_User1, 'ro-', ...
   'LineWidth',1.5, 'MarkerSize',7);
hold on;

semilogy(SNR_dB, ber_User2, 'bs-', ...
   'LineWidth',1.5, 'MarkerSize',7);

semilogy(SNR_dB, ber_OFDMA, 'k--', ...
   'LineWidth',1.5);

grid on;

xlabel('SNR (dB)');
ylabel('Bit Error Rate (BER)');

title('NOMA Power-Domain Transceiver Performance');

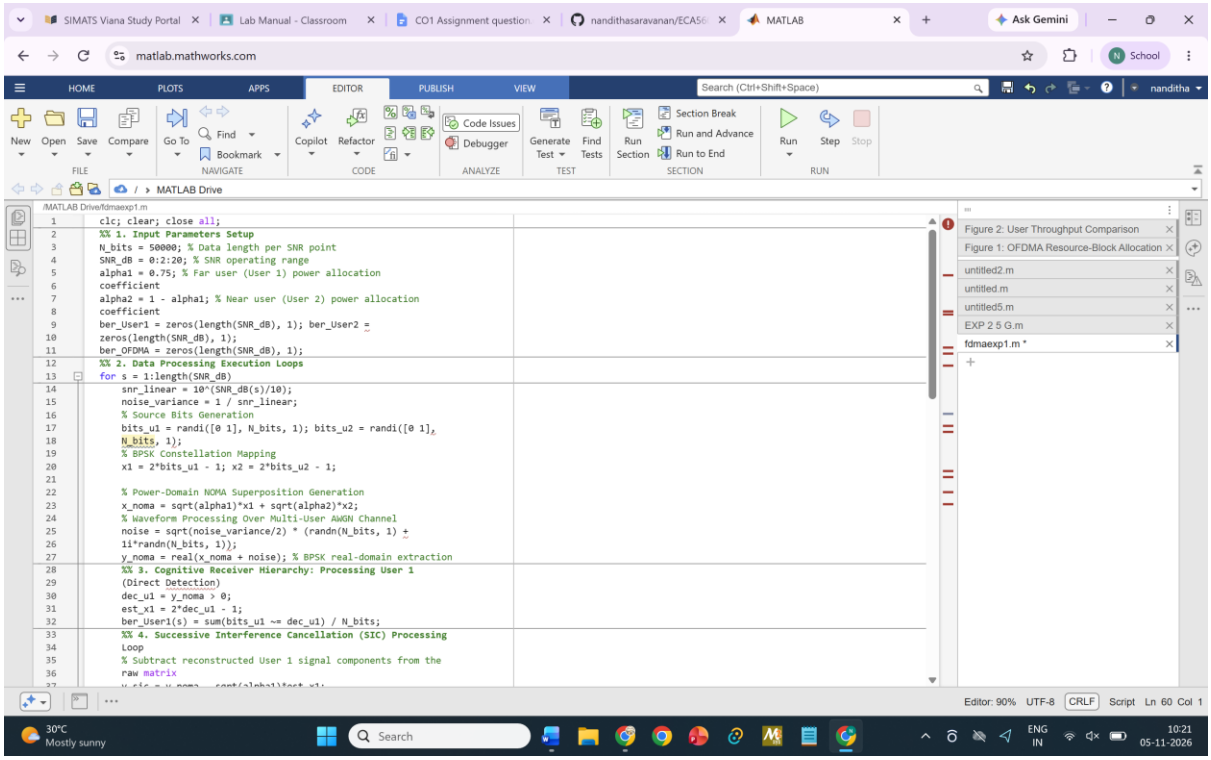
legend('User 1 (Far User)', ...
     'User 2 (Near User after SIC)', ...
     'OFDMA Benchmark', ...
     'Location','best');

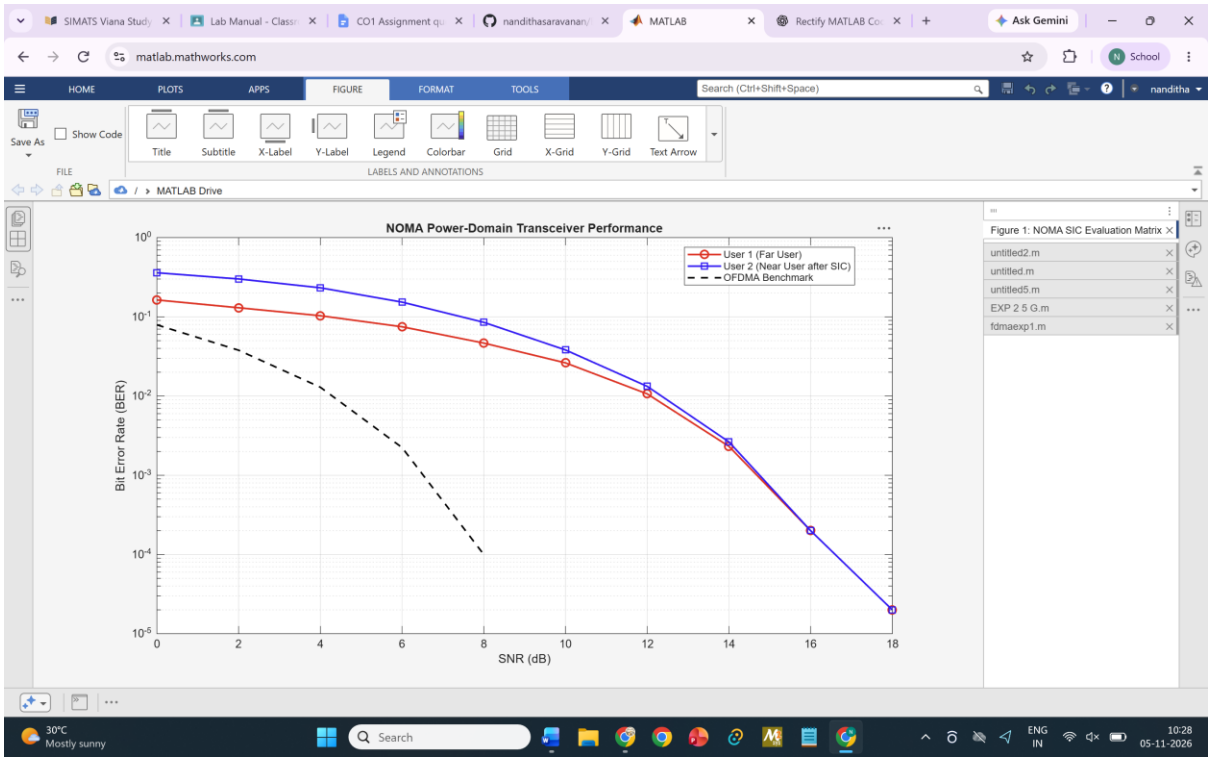
ylim([1e-5 1]);
```

```
hold off;


%% 7. Display BER Results
disp('SNR (dB)    User 1 BER    User 2 BER    OFDMA BER');


for s = 1:length(SNR_dB)
   fprintf('%5.1f      %10.6f    %10.6f    %10.6f\n', ...
      SNR_dB(s), ber_User1(s), ber_User2(s), ber_OFDMA(s));
endOutput:
```




NOMA Power-Domain Transceiver Performance
User 1 (Far User)
User 2 (Near User after SIC)
OFDMA Benchmark
Bit Error Rate (BER)
SNR (dB)
Figure 1: NOMA SIC Evaluation Matrix
untitled2.m
untitled.m
untitled5.m
EXP 2 5 G.m
fdmaexp1.m
MATLAB Drive
matlab.mathworks.com
30°C
Mostly sunny
10:28
05-11-2026